

# Knowledge Distillation

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## What Is It?

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**Knowledge Distillation (KD)** is a technique for transferring knowledge from a larger, stronger model, the **teacher**, into a smaller, faster model, the **student**.

Instead of training the student only on hard ground-truth labels, the student also learns to imitate the teacher's behavior: its output probabilities, its generated answers, or its reasoning steps.

## A Simple Example

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Suppose the task is next-token prediction, and the true answer is "Paris." A normal label only tells the student one fact:

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Hard label:  
Correct token = Paris
```

The teacher, however, produces a full probability distribution:

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Teacher logits (raw):  
Paris → 0.80  
London → 0.10  
Berlin → 0.05  
Rome → 0.05
```

This distribution tells the student more than a single correct answer: Paris is very likely, London is somewhat plausible, and Berlin/Rome are unlikely but not impossible. This extra information is often called **dark knowledge**.

## Training Objective

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A common distillation objective combines normal supervised learning with teacher imitation:

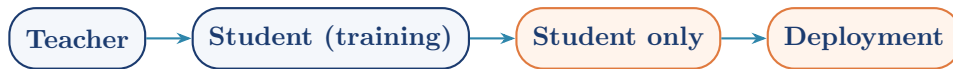
$$\mathcal{L} = \alpha \mathcal{L}_{\text{normal}} + (1 - \alpha) \mathcal{L}_{\text{distillation}},$$

- $\mathcal{L}_{\text{normal}}$ : compares the student's prediction with the ground-truth label.
- $\mathcal{L}_{\text{distillation}}$ : compares the student's distribution with the teacher's, typically using **KL-divergence**:  $D_{\text{KL}}(p_{\text{teacher}} \parallel p_{\text{student}})$ .
- $\alpha$ : controls the balance between the two.

## Training vs. Deployment

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The main practical payoff of KD: the teacher is only needed during training. At inference time, only the (cheap) student is served.



## Types of Knowledge Distillation

For LLMs, the teacher’s “signal” can take several forms.

### 1. Logit Distillation

#### Match output probabilities

The student is trained so that  $p_{\text{student}}(y | x) \approx p_{\text{teacher}}(y | x)$  over the full vocabulary, not just the top answer.

| Token | Teacher | Student |
|-------|---------|---------|
| A     | 0.70    | 0.65    |
| B     | 0.20    | 0.23    |
| C     | 0.10    | 0.12    |

### 2. Response Distillation

#### Match full generated answers

Prompt: Explain why the sky is blue.

Teacher Answer: The sky appears blue because shorter wavelengths of sunlight are scattered more strongly by the atmosphere.

The pair (Prompt, Teacher Answer) becomes a training example for the student.

### 3. Reasoning Distillation

#### Match intermediate reasoning steps

Question: John has 5 apples and buys 3 more. How many does he have?

Teacher:  $5 + 3 = 8$ . Answer: 8

Useful for math, coding, and multi-step reasoning tasks.

### 4. Feature Distillation (less common, but worth knowing)

Instead of matching only the final output, the student is trained to match the teacher’s **intermediate hidden states or attention maps**. Used in models like DistilBERT.

## Pseudo-Labels

When unlabeled data is abundant, the teacher can label it automatically:

$$x \xrightarrow{\text{teacher}} \hat{y}_T \implies (x, \hat{y}_T) \text{ becomes training data}$$



## When Is Knowledge Distillation Plausible?

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### A strong teacher already exists

Distillation needs a good teacher. If no strong model is already available or accessible, there is nothing to distill.

### Serving the teacher directly is too expensive

Large Teacher (multi-GPU) → Small Student (single GPU)

### The student has enough capacity

70B → 7B often preserves substantial knowledge.  
70B → *extremely tiny model* often causes a large drop, this is called the **capacity gap** problem.

### Plenty of (unlabeled) data is available

Unlabeled inputs → teacher-generated pseudo-labels → training signal for the student.